Daughters' Days: Son Preference, Gender Norms and the Time Use of Sisters and Brothers in India

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Abstract

Son preference in India is documented in sex ratios, mortality, nutrition and health care, but much less in how households allocate children's own time. Using 24-hour diaries from India's Time Use Survey 2024 and sibling-set fixed effects, which absorb everything siblings share, I compare sisters and brothers aged 6 to 17. Among 34,133 children in 13,984 mixed-sex sets, girls do 40.9 more minutes of unpaid domestic work a day than their brothers (SE 1.0) and have 33.8 fewer minutes of free time (SE 1.2). The domestic-work gap widens with age, so the gendered division of household labour takes shape before most girls marry. It is smaller but remains large and significant in two-child sets and in sets whose eldest child is a boy, so fertility-stopping selection can explain at most part of it. Unequal treatment of daughters within Indian households extends from parental inputs to the allocation of their own time.

Keywords: son preference, child time use, siblings, domestic work, gender norms, India
JEL codes: J13, J16, O15, I25, D13

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1 Introduction

Son preference is among the most extensively documented forms of gender discrimination in India. It appears in the deficit of women in the population (Sen 1990; Anderson and Ray 2010), in male-biased sex ratios at birth (Jha et al. 2006; Guilmoto 2009), in excess mortality of girls (Das Gupta 1987; Rose 1999), in fertility behaviour that depends on the sex of the children already born (Arnold et al. 1998; Clark 2000; Jayachandran 2017), and in parental investments in girls' nutrition, immunisation and health care (Behrman 1988; Pande 2003; Mishra et al. 2004; Oster 2009). Within the household, girls are breastfed for shorter periods (Jayachandran and Kuziemko 2011), receive less childcare time and fewer health inputs than boys (Barcellos et al. 2014), and are on average shorter than their brothers in ways that depend on birth order (Jayachandran and Pande 2017).

This literature asks what parents give to sons and to daughters. A second margin is what parents ask of them. In a household economy in which cooking, cleaning, fetching water and minding small children are women's work, daughters may be drawn into that work long before adulthood. If so, discrimination appears not only in the inputs that girls receive but in the hours they have left for school, study, play and rest. These hours matter for the same reasons that nutrition and health inputs matter. Time spent in domestic work during school ages competes with learning (Assaad et al. 2010), and the tasks a child is assigned are one way in which norms about who does housework are passed from one generation to the next (Akerlof and Kranton 2000; Bisin and Verdier 2001; Dhar et al. 2019).

Measuring this margin requires diaries for children themselves. Child-labour surveys typically record whether a child worked and, sometimes, hours in market or family-enterprise work, while domestic chores are often recorded separately or not at all (Edmonds 2008). Surveys of parental investment record inputs rather than the child's day. India's Time Use Survey 2024 (National Statistics Office 2025) records a 24-hour diary for every household member aged six and above, coded to the International Classification of Activities for Time-Use Statistics (United Nations Statistics Division 2016). In 100% of the sibling sets studied here, sisters and brothers report the same diary day, so siblings can be compared directly on the same day.

I compare girls and boys aged 6 to 17 within the same sibling set: the never-married children of the household head, or the head's grandchildren, who live in the same household. A sibling-set fixed effect absorbs everything the children share: household income and assets, caste and religion, location, the parents' preferences and bargaining position, the household's demand for domestic work, and, in almost all sets, the diary day itself. What remains is the difference between sisters and brothers of the same single-year age in the same household. This design removes the main objection to cross-sectional comparisons of girls and boys, namely that girls and boys live in different kinds of households because son preference shapes fertility (Yamaguchi 1989; Basu and de Jong 2010).

The fixed effect does not remove all selection. Which children are born, survive and live at home is itself shaped by son preference. Families that stop having children once they have a son produce

mixed-sex sets in which daughters are, on average, older than their brothers (Yamaguchi 1989; Kugler and Kumar 2017). If older children do more household work for reasons unrelated to sex (Edmonds 2006; Emerson and Souza 2008), a within-set comparison could confound sex with birth order. I address this in four ways: by controlling for age rank within the set, by estimating eldest-daughter effects directly, by restricting the sample to two-child sets, and by comparing sets whose eldest child is a boy, where the stopping rule after a son cannot have produced the girl-first ordering, with sets whose eldest child is a girl.

The estimates are as follows. Within mixed-sex sibling sets, girls do 40.9 minutes (SE 1.0) more unpaid domestic work per day than their brothers and 1.6 minutes (SE 0.2) more caregiving. They spend 2.6 minutes (SE 1.5) more time in learning and have 33.8 minutes (SE 1.2) less free time. Brothers' mean domestic work is 5 minutes a day. The gap is therefore large relative to the base: it is 8.8 times their brothers' mean. By age, the domestic-work gap is 4.3 minutes (SE 1.0) at ages 6 to 10 and 104.7 minutes (SE 2.7) at ages 15 to 17. The gendered division of domestic labour thus widens through adolescence, well before marriage. The domestic-work gap in two-child sets and in sets with an elder brother is smaller than the baseline but remains large and significant, so fertility stopping can account for at most part of it. For mothers, I find a significant association between having only daughters among co-resident children aged 6 to 17 and active childcare; the difference is 2.7 minutes (SE 0.8).

The paper makes three contributions. First, it adds children's own time to the evidence on son preference and intrahousehold discrimination in India, which has focused on survival, nutrition, health and education spending (Das Gupta 1987; Behrman 1988; Kingdon 2005; Barcellos et al. 2014; Azam and Kingdon 2013). Time is an input whose allocation is visible in a single diary and that does not depend on prices or on the assignment of shared goods, which has made expenditure-based tests of discrimination hard to interpret (Deaton 1989). Second, it measures the full day, so that any gap in domestic work can be traced to the activities it displaces. Because each diary sums to 1,440 minutes, the within-sibling gaps across all nine ICATUS divisions sum to zero, and the paper reports the complete reallocation rather than a single outcome. Third, it documents when the adult gender gap in unpaid work (Hirway and Jose 2011) emerges, by single-year age and by birth order within the family, which bears on the formation of gender norms in adolescence (Dhar et al. 2022; Beaman et al. 2012).

The rest of the paper is organised as follows. Section 2 reviews the evidence on son preference, children's work, adolescent norms and sibling designs. Section 3 describes the data and the construction of sibling sets. Section 4 sets out the empirical strategy and the threats to it. Section 5 reports the main results, the reallocation of the day, the age profile, birth order and heterogeneity. Section 6 examines fertility-stopping selection and other robustness checks. Section 7 turns to mothers' time. Section 8 discusses interpretation and policy, and Section 9 concludes.

2 Background and related literature

2.1 Son preference: sex ratios, mortality and fertility stopping

Sen (1990) estimated that, given the sex ratios found where women and men receive similar care, more than 100 million women were missing from the populations of Asia and North Africa, with India a large contributor. Anderson and Ray (2010) show that excess female mortality in India is not confined to birth and early childhood but extends into adult ages. Within India, the deficit of girls has long been concentrated in the north-west. Miller (1981) documented the neglect of female children in rural north India, and Dyson and Moore (1983) related the contrast between north and south to differences in kinship systems and in women's autonomy. Das Gupta et al. (2003) argue that son preference in China, India and the Republic of Korea is rooted in patrilineal and patrilocal family systems in which sons stay with and support their parents while daughters marry out.

Son preference operates through fertility as well as through survival. Arnold et al. (1998) show that the sex composition of surviving children shapes the family-building process in India, and Clark (2000) that couples' fertility depends on the number of sons they already have. When desired family size falls, the conflict between having few children and having a son intensifies, and sex selection rises (Jayachandran 2017; Guilmoto 2009). Jha et al. (2006) find that the sex ratio of second and third births in India is heavily male-biased when the earlier children were girls, but not when they were boys. Policy can shift the trade-off: Anukriti (2018) shows that financial incentives in Haryana that rewarded small families lowered fertility while raising the share of sons. The spread of prenatal sex determination changed which girls are born, and Hu and Schlosser (2015) find improvements in the nutritional status of girls relative to boys as it spread. The cost of son preference also falls on mothers: Milazzo (2018) links son-biased fertility behaviour to shorter birth spacing and higher maternal mortality, and Rosenblum (2013) links excess female child mortality to parents' fertility decisions.

2.2 Inputs to girls: nutrition, health, breastfeeding and birth order

A second strand looks inside the household. Rosenzweig and Schultz (1982) show that the survival advantage of boys over girls in rural India is smaller where women's expected labour market returns are higher. Behrman (1988) finds that boys are favoured in the allocation of nutrients in the lean season. Das Gupta (1987) shows that in rural Punjab the excess mortality of girls is concentrated among girls born to mothers who already have daughters. Pande (2003) finds that girls' nutrition and immunisation depend on the sex composition of their older siblings, and Mishra et al. (2004) document sex differentials in feeding, health care and nutritional status across India. Jayachandran and Kuziemko (2011) show that girls are breastfed for shorter periods because mothers who have not yet had the desired number of sons stop breastfeeding sooner in order to conceive again. Barcellos et al. (2014) find that boys receive more childcare time and more health inputs than girls. Oster (2009) shows that the size of gender gaps in children's health investments depends on the cost of

access to care. Jayachandran and Pande (2017) show that the height disadvantage of Indian children relative to children in sub-Saharan Africa emerges with second and later births, and that eldest sons in India have a height advantage, which they relate to son preference and the parental resources concentrated on the eldest son.

Expenditure-based tests of discrimination have been harder to interpret. Deaton (1989) proposed testing whether the arrival of a boy or a girl reduces spending on adult goods by different amounts; such tests often fail to detect discrimination that appears in mortality and health. For education, Kingdon (2002) documents a large gender gap in educational attainment in India and asks how much of it observed characteristics can explain, and Kingdon (2005) and Azam and Kingdon (2013) find evidence of pro-boy bias in household spending on education, concentrated in particular states and age groups. Policy that lowers the cost of attending school can narrow the gender gap: Muralidharan and Prakash (2017) show that a programme providing bicycles to girls in Bihar raised girls' secondary-school enrolment.

Two features of this literature matter for the present paper. First, gaps in parental inputs reflect both preferences and the returns that parents expect from investing in sons and daughters (Rosenzweig and Schultz 1982; Pitt et al. 1990). A gender gap in children's time is subject to the same ambiguity. Second, the sex composition of siblings and birth order shape outcomes, which motivates a design that holds the family fixed and examines birth order explicitly.

2.3 Children's work, domestic chores and schooling

The economics of child labour has concentrated on work for pay and in family enterprises. Basu and Van (1998) model households that send children to work only when adult income is too low, and Basu (1999) and Edmonds (2008) review the theory and evidence. Edmonds (2008) emphasises that most working children work for their own households, and that domestic chores are often left out of child-labour measures even though they take a large share of children's time, particularly girls'. Webbink et al. (2012) compare housework and family-business work of children across 16 developing countries and find that girls do more housework than boys. Assaad et al. (2010) show for Egypt that girls' domestic work reduces their schooling and that measures of child labour that exclude domestic work understate girls' work.

Sibling differences in child work depend on birth order and sex. Edmonds (2006) uses data from Nepal to study why siblings differ in child labour, including by birth order and sex. Emerson and Souza (2008) find for Brazil that earlier-born children are more likely to work and less likely to attend school. Parish and Willis (1993) argue that in Taiwan older daughters' early entry into work helped finance their younger siblings' schooling. In India, evidence on children's domestic work comes mainly from participation questions in household surveys and from the official time-use reports (National Statistical Office 2020; National Statistics Office 2025), which report averages by sex and age but do not compare children within the same household. Hirway and Jose (2011) use Indian time-use data to show how much of women's work is unpaid and unrecorded in conventional

labour statistics.

2.4 Adolescence and the formation of gender norms

Gender norms about housework are learned in the family (Akerlof and Kranton 2000), and Indian adolescents' gender attitudes track their parents' (Dhar et al. 2019) but can be shifted by school programmes (Dhar et al. 2022) and by exposure to female leaders, which also reduced girls' time in household chores (Beaman et al. 2012). Marriage is the point at which many girls leave school (Field and Ambrus 2008; Jensen 2012). Daughters' domestic work in adolescence is one channel through which these norms take effect (Jayachandran 2015).

2.5 Sibling-composition designs and fertility-stopping selection

The sex composition of siblings has been used both as a source of variation and as an outcome. Butcher and Case (1994) find that women raised with only brothers attain more education, while Kaestner (1997) revisits the evidence and finds little effect. Garg and Morduch (1998) and Morduch (2000) show that, in Ghana and other African countries, children's health and schooling depend on whether their siblings are sisters or brothers, which they interpret as competition for household resources. Angrist and Evans (1998) use the sex mix of the first two children as a source of exogenous variation in family size in the United States, and Dahl and Moretti (2008) use the sex of the first-born child to measure parents' demand for sons. Black et al. (2005) show that birth order, rather than family size, matters for children's education in Norway. Lundberg (2005) reviews the evidence on how parents' behaviour depends on the sex of their children.

The validity of such designs rests on the sex of a child being as good as random. In India it is not. Son-preferring stopping rules imply that girls have more siblings on average than boys and that sons tend to be later-born (Yamaguchi 1989; Basu and de Jong 2010), and sex selection after the first birth changes which children are born (Jha et al. 2006). Kugler and Kumar (2017) use the sex of the first-born child as an instrument for family size in India, on the grounds that the sex of the first-born child is close to random, and find that larger families reduce children's education. The design in this paper uses within-family variation, but which children co-reside is still selected. Section 4 discusses the implications, and Section 6 reports tests.

3 Data and measurement

3.1 The Time Use Survey 2024

The Time Use Survey 2024 was conducted by the National Statistics Office from January to December 2024 (National Statistics Office 2025). It follows the 2019 round (National Statistical Office 2020) and covers a nationally representative sample of households. Every household member aged six and above reports a diary of the 24 hours from 4 a.m. on the day before the interview. Activities are coded at three digits to the ICATUS 2016 classification (United Nations Statistics

Division 2016). Where more than one activity is recorded in a time interval, one is flagged as the major activity. I use every activity record in the file (10,211,478 records, 450,457 diaries in 139,487 households and 9,969 first-stage units), with no sampling of records. Minutes are computed from major activities only, so each diary should sum to 1,440 minutes; 100.0% of diaries satisfy this exactly. All estimates use the survey weights.

The survey asks who reported each diary. Children's diaries are often reported by a parent. In the analysis sample, 69.6% of diaries (weighted) were reported by another household member. I report results restricted to self-reported diaries in Section 6.

3.2 Children, sibling sets and mothers

The analysis sample is household members aged 6 to 17 who are never married and are either unmarried children of the head or grandchildren of the head. There are 82,223 such children. A sibling set is the children with the same relationship code in the same household: the head's children form one set and the head's grandchildren another. Grandchildren in a household with more than one married child of the head may be cousins rather than siblings; I report results for the head's children only in Section 6. Of all children in the sample, 66.3% live in a set with at least one other child aged 6 to 17 and 41.5% in a set with at least one girl and one boy. The fixed-effects sample is the 34,133 children in the 13,984 mixed-sex sets, with a mean set size of 2.4 children; 15.1% of these sets are grandchildren of the head.

To study heterogeneity by parental education and to examine mothers' time, I link each set to its mother. For the head's children, the mother is the unique woman aged 15 or older among the head and the head's spouse. For grandchildren, she is the unique woman among the head's married children and their spouses. Links are dropped when the age difference between mother and child is below 15 or above 50 years. A mother is linked for 93.3% of children in the fixed-effects sample. Where no unique mother can be identified, the set remains in the main analysis but drops out of the parental-education interaction. I also record the education of the household head.

Within each set, I rank children by age from the eldest (rank 1). A child is the eldest if she or he is strictly older than all other children aged 6 to 17 in the set. Because only co-resident children aged 6 to 17 are observed, the eldest child in the data need not be the first-born: older siblings may have left home, married, or be aged 18 or more. I therefore also count co-resident siblings aged 18 or older and report estimates for sets that have none.

3.3 Outcomes

The outcomes are minutes per day in ICATUS activity groups defined from the three-digit codes. Unpaid domestic work is division 3 (food and meals, cleaning and upkeep of the dwelling, care of textiles, shopping, household management, pet care, and related travel). Unpaid caregiving is division 4, most of which for children in this age range is care for younger siblings. Learning is

division 6, which includes attending school, homework and study. Production work is division 1 (employment) and division 2 (production of goods for own final use, including fetching water and firewood). Free time is divisions 7 and 8 (socialising, community participation, religious practice, culture, leisure, mass media and sports). Self-care is division 9, which includes sleep. I also report participation on the diary day and two indicators from the household roster: whether the child's usual principal activity status is attending an educational institution, and whether it is domestic duties.

Table 1: Children aged 6–17 in mixed-sex sibling sets: minutes per day and characteristics

Outcome	Boys	Girls
Unpaid domestic work	4.7 (0.2)	48.5 (1.0)
Unpaid caregiving (incl. siblings)	2.2 (0.2)	4.1 (0.2)
Learning (school, homework, study)	367.1 (2.7)	368.9 (2.7)
Paid and own-use production work	20.5 (1.0)	12.1 (0.7)
Free time	286.5 (2.4)	250.8 (2.2)
Sleep	582.2 (1.1)	578.9 (1.2)
Self-care incl. sleep	758.5 (1.2)	755.2 (1.3)
Age (years)	12.0 (0.0)	12.5 (0.0)
Children aged 6–17 in set	2.6 (0.0)	2.7 (0.0)
Any domestic work (share)	0.06 (0.00)	0.34 (0.00)
Attending education (share)	0.94 (0.00)	0.93 (0.00)
Domestic duties (share)	0.01 (0.00)	0.06 (0.00)
Eldest in set (share)	0.37 (0.00)	0.43 (0.00)
Proxy-reported diary (share)	0.71 (0.01)	0.68 (0.01)

Notes: Survey-weighted means; FSU-clustered standard errors in parentheses. Minutes per day unless marked as a share.

Table 1 reports means for girls and boys in the fixed-effects sample. These raw means compare all girls with all boys in mixed sets and do not adjust for age; the within-set estimates below do.

4 Empirical strategy

4.1 Within-sibling comparison

For child i in sibling set j, I estimate

$$y_{ij} = \beta\, Girl_{ij} + \alpha_{a(ij)} + \lambda_j + \varepsilon_{ij},$$

where $\alpha_{a(ij)}$ are single-year-of-age effects and λ_j is a sibling-set fixed effect. The coefficient β is the average difference between a girl and a boy of the same age in the same set. The fixed effect absorbs household income, consumption, land and assets, caste and religion, state and rural or urban location, the number of household members and their ages, the parents' education and preferences, and the season of the survey. It also absorbs the diary day for sets in which all siblings report the same day, which is 100% of mixed-sex sets; I report estimates with child-level day-of-week indicators as a check.

For comparison, I estimate the same regression on all children aged 6 to 17, including only children and single-sex sets, without the set fixed effect and with controls for single-year age, rural or urban residence, consumption quintile, household size, the number of members under six, caste, religion, whether the diary was proxy-reported, and state and day-of-week fixed effects. Under son-biased fertility stopping, girls and boys live in different kinds of households, so this cross-sectional estimate need not equal the within-set estimate. Comparing the two shows how much the household-level selection matters.

To study when the gap emerges, I interact $Girl_{ij}$ with three age groups (6 to 10, 11 to 14 and 15 to 17) in a single regression on all mixed-sex sets. This uses every set, including those in which sisters and brothers fall in different age groups. To study birth order, I add an indicator for being the eldest child in the set and its interaction with $Girl_{ij}$. With single-year age effects and set fixed effects, the eldest indicator is identified from differences in age gaps across sets. The interaction measures how much more (or less) an eldest daughter's time differs from an eldest son's than a younger daughter's differs from a younger son's. For heterogeneity by household characteristics X_j, which are constant within a set, I estimate $\beta + \gamma X_j$ by adding $Girl_{ij} \times X_j$.

Because each diary sums to 1,440 minutes and the estimator is linear, the coefficients β for the nine ICATUS divisions sum to zero. In the sample they sum to zero to two decimal places. This identity makes the reallocation of time explicit: any extra minutes of domestic work must come from another activity.

4.2 Threats to identification

Fertility stopping and birth order. Under a son-preferring stopping rule, parents who have a son stop having children, and those who do not continue. Mixed-sex sets then disproportionately consist of one or more older girls followed by a younger boy (Yamaguchi 1989; Basu and de Jong 2010). In the fixed-effects sample, 54.9% of sets whose eldest child is uniquely identified have a girl as the eldest. If eldest children are assigned more domestic work, then β mixes the effect of sex with that of birth order. Single-year age effects remove differences in age but not in rank. I add age-rank indicators, estimate the girl-by-eldest interaction, and estimate β separately in sets whose eldest child is a boy, where the girl in the set is younger than at least one brother and cannot be the product of continued childbearing in pursuit of a first son. I also estimate β in two-child sets, where the distribution of birth order is simplest. None of these restrictions creates random assignment of sex, but if the gap reflects sex rather than position in the family, it should survive all of them.

Selection on co-residence. Girls in India marry earlier than boys, and a girl who marries before 18 leaves the sample. Girls who remain at home at ages 15 to 17 may differ from those who left. Daughters may also be sent to live with relatives to help with housework, and sons may be sent away to study. The data do not record children who live elsewhere. Section 6 reports estimates for sets with no co-resident siblings aged 18 or older, which are less likely to have lost members to marriage or migration.

Proxy reporting. If parents who report their children's diaries describe sons' and daughters' days according to gender scripts, measured gaps could overstate real ones. I estimate β on self-reported diaries only.

The diary day. A single day may be a school day for one child and not for another, for example if siblings attend schools with different holidays. Schools in India are generally closed on Sundays, so I estimate the gap separately for diaries recorded on Sundays and on other days.

4.3 Inference

Standard errors are clustered on the first-stage unit (FSU), the primary sampling unit, which nests the household and the sibling set (Cameron and Miller 2015; Abadie et al. 2023). The fixed-effects sample spans 6,695 FSUs. I weight by the survey weights and report unweighted estimates as a check (Solon et al. 2015). The outcomes in the main table share one diary and are mechanically related, so I interpret them jointly through the time-accounting identity rather than as independent tests.

5 Results

5.1 Sisters and brothers

Table 2: Girl minus boy, children aged 6–17

Outcome	Within siblings	Within siblings + rank	All children, controls	Boys' mean
A. Minutes per day				
Unpaid domestic work	40.9*** (1.0)	40.7*** (1.0)	34.9*** (0.6)	4.7
Unpaid caregiving (incl. siblings)	1.6*** (0.2)	1.6*** (0.2)	1.6*** (0.2)	2.2
Learning (school, homework, study)	2.6* (1.5)	2.9** (1.5)	5.6*** (1.6)	367.1
Paid and own-use production work	-11.2*** (1.0)	-11.5*** (1.0)	-10.6*** (0.7)	20.5
Free time	-33.8*** (1.2)	-33.7*** (1.2)	-31.7*** (1.3)	286.5
Sleep	0.0 (0.7)	0.2 (0.7)	-0.2 (0.7)	582.2
B. Percentage points				
Any domestic work on diary day	26.3*** (0.5)	26.2*** (0.5)	22.2*** (0.3)	6.1
Any caregiving on diary day	1.6*** (0.2)	1.6*** (0.2)	1.8*** (0.2)	2.4
Any learning on diary day	0.5 (0.3)	0.6* (0.3)	1.0*** (0.3)	85.3
Usual status: attending education	-0.4 (0.3)	-0.4 (0.3)	-0.4* (0.2)	93.9
Usual status: domestic duties	4.4*** (0.2)	4.3*** (0.2)	4.1*** (0.2)	1.0

Notes: Within siblings: sibling-set fixed effects and single-year age effects; 34,133 children in 13,984 mixed-sex sets. Within siblings + rank: adds indicators for age rank in the set (1, 2, 3 or higher). All children: 82,223 children, with single-year age, urban, consumption quintile, household size, members under six, caste, religion, proxy report, and state and day-of-week fixed effects. Boys' mean: boys in mixed-sex sets. Panel B outcomes are shares, reported in percentage points. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 2 reports the main results. Within sibling sets, girls do 40.9 minutes (SE 1.0) more unpaid domestic work than their brothers of the same age and 1.6 minutes (SE 0.2) more caregiving. They spend 2.6 minutes (SE 1.5) more time in learning, and production work, paid or for own use, differs by -11.2 minutes (SE 1.0). Free time differs by -33.8 minutes (SE 1.2) and sleep by 0.0 minutes (SE

0.7). Taken together, girls do 42 more minutes of unpaid domestic and care work a day than their brothers, about 5.0 hours a week.

The extensive margins in panel B show how the gaps arise. The probability of doing any domestic work on the diary day is 26.3 percentage points (SE 0.5) higher for girls than for their brothers, and the probability of any caregiving is 1.6 percentage points (SE 0.2) higher. The probability of recording any learning on the diary day is 0.5 percentage points (SE 0.3) higher. In the household roster, the probability that a girl's usual principal activity is attending an educational institution is 0.4 percentage points (SE 0.3) lower than her brother's, and the probability that it is domestic duties is 4.4 percentage points (SE 0.2) higher. The roster does not show a significant within-family gap in school attendance, so any gap in learning time arises mainly among enrolled children.

Adding age-rank indicators changes the domestic-work gap from 40.9 to 40.7 minutes. The gap is not driven by girls' position in the birth order. The cross-sectional estimate for all children, with household controls, is 34.9 minutes (SE 0.6) for domestic work and 5.6 minutes (SE 1.6) for learning. The difference between the cross-sectional and within-set estimates shows that girls and boys live in different kinds of households, as fertility stopping predicts, and that controls for observed household characteristics do not remove this difference.

5.2 Where the minutes come from

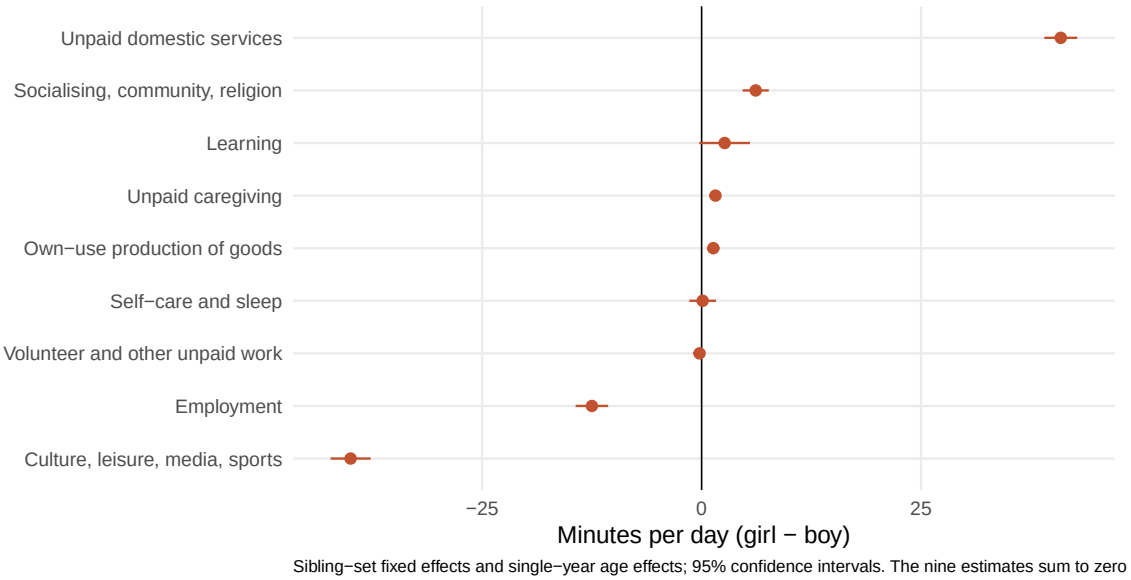


Figure 1: Girl minus boy within sibling sets, by ICATUS division (minutes per day)

Figure 1 reports the within-sibling gap for each of the nine ICATUS divisions. The largest offsetting difference is in culture, leisure, media, sports (-40.0 minutes, SE 1.2), followed by employment (-12.5 minutes, SE 0.9). Girls' extra domestic time does not come mainly out of learning. It also comes out of leisure, which matters for children's well-being and for the play and study that leisure time

can contain. The identity that the nine differences sum to zero means that no gap can be read in isolation: a finding that girls do more housework is also a finding about which other activities they do less of.

5.3 Which tasks

Table 3: Domestic and care tasks: girl minus boy, minutes per day

Task	Girl minus boy, within siblings	Boys' mean	Girls' mean
Food and meals	27.0*** (0.7)	0.6	29.5
Cleaning and upkeep of dwelling	8.4*** (0.3)	0.6	9.4
Care of textiles (laundry)	4.7*** (0.2)	0.3	5.3
Shopping	-0.7*** (0.1)	1.5	0.9
Other domestic services	1.5*** (0.2)	1.7	3.4
Childcare (mostly younger siblings)	1.3*** (0.2)	2.1	3.7
Care of adults and other care	0.3*** (0.1)	0.2	0.4
Fetching water and firewood	1.6*** (0.1)	0.7	2.4

Notes: Sibling-set fixed effects and single-year age effects. The first five rows partition unpaid domestic services (ICATUS division 3) and the next two partition unpaid caregiving (division 4). Fetching water and firewood is own-use production (division 2). FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 3 breaks the domestic and care gaps into tasks. The largest single component of the domestic-work gap is food and meals (27.0 minutes, SE 0.7). Childcare differs by 1.3 minutes (SE 0.2), and fetching water and firewood by 1.6 minutes (SE 0.1). Girls' extra work is concentrated in the routine, daily tasks of cooking and cleaning, which must be done at fixed times and are hard to combine with a school timetable. Shopping, which takes a child outside the home, differs by -0.7 minutes (SE 0.1).

5.4 Age

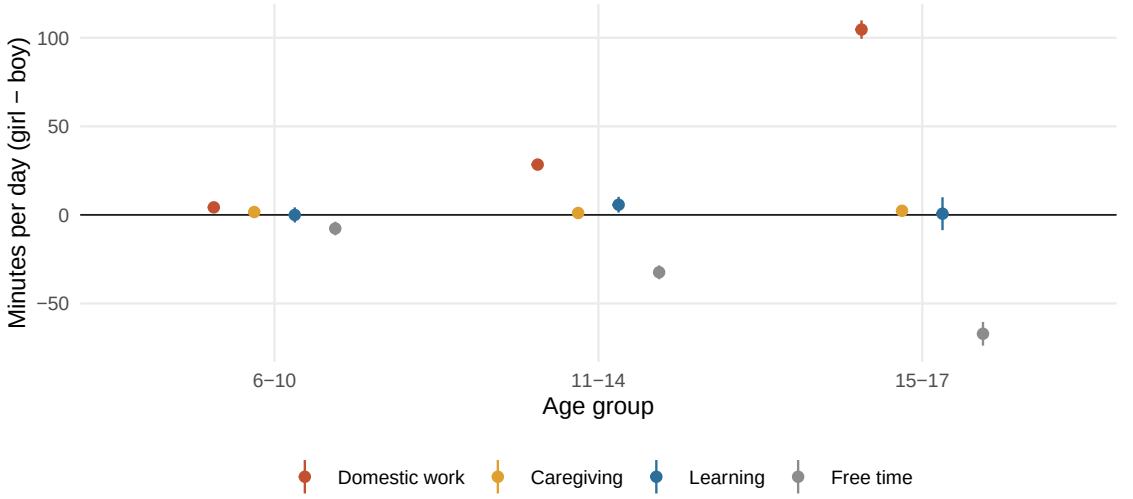


Figure 2: Girl minus boy within sibling sets, by age group

Figure 2 shows the gaps by age group, estimated in one regression per outcome on all mixed-sex sets. The domestic-work gap is 4.3 minutes (SE 1.0) at ages 6 to 10, 28.4 minutes (SE 1.4) at ages 11 to 14 and 104.7 minutes (SE 2.7) at ages 15 to 17. The difference between the oldest and youngest groups is 100.4 minutes (SE 2.8). The gap opens through adolescence. By ages 15 to 17, the within-family gap equals 35% of the raw gap in domestic work between women and men aged 25 to 59 (297 minutes). For learning, the gap is -0.0 minutes (SE 2.2) at ages 6 to 10 and 0.7 minutes (SE 4.7) at ages 15 to 17 (difference 0.7 minutes, SE 5.2). The learning gap does not widen significantly with age.

5.5 Birth order and eldest daughters

Table 4: Birth order within the sibling set: minutes per day

	Domestic	Care	Learning	Free time
Girl (not eldest)	13.6*** (1.2)	0.4 (0.3)	3.2 (2.2)	-18.1*** (1.9)
Eldest (boys)	-33.7*** (1.8)	-1.8*** (0.6)	8.2** (3.7)	17.7*** (3.0)
Girl × eldest	67.2*** (2.6)	2.8*** (0.7)	-1.1 (5.1)	-38.8*** (4.1)
Eldest daughter vs eldest son	80.8*** (2.1)	3.2*** (0.5)	2.1 (3.7)	-56.9*** (2.9)

Notes: Each column is one regression on mixed-sex sets with sibling-set fixed effects, single-year age effects, an indicator for the eldest child in the set and its interaction with girl. The last row is the sum of the first and third rows (the girl-boy gap among eldest children). FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 4 examines birth order. Among children who are not the eldest, girls do 13.6 minutes (SE 1.2) more domestic work than boys. Being the eldest changes a boy's domestic work by -33.7 minutes (SE 1.8), and the interaction of girl and eldest is 67.2 minutes (SE 2.6). An eldest daughter does 80.8

minutes (SE 2.1) more domestic work than an eldest son. The eldest daughter carries an extra share of the household's domestic work, over and above the gap between younger sisters and brothers. This pattern is consistent with eldest daughters substituting for their mothers in household work. For caregiving, the girl-by-eldest interaction is 2.8 minutes (SE 0.7), and for learning -1.1 minutes (SE 5.1). Eldest daughters in particular look after their younger siblings.

5.6 Heterogeneity

Table 5: Heterogeneity of the girl-boy gap: minutes per day

	Domestic	Care	Learning	Free time
Urban household (N = 34,133)				
Girl	44.6*** (1.1)	1.8*** (0.3)	0.5 (1.7)	-35.5*** (1.4)
Girl × moderator	-16.1*** (2.2)	-0.8* (0.5)	9.2*** (3.5)	7.3** (3.0)
Top two consumption quintiles (N = 34,133)				
Girl	43.8*** (1.1)	2.1*** (0.3)	2.1 (1.7)	-35.2*** (1.4)
Girl × moderator	-11.1*** (2.0)	-1.9*** (0.4)	1.9 (3.3)	5.2* (2.9)
Mother has secondary education or more (N = 31,832)				
Girl	48.8*** (1.2)	1.7*** (0.3)	0.2 (1.9)	-37.0*** (1.5)
Girl × moderator	-32.2*** (1.6)	-1.1*** (0.4)	12.4*** (2.9)	11.8*** (2.5)
Head has secondary education or more (N = 33,911)				
Girl	47.5*** (1.2)	1.8*** (0.3)	0.4 (1.9)	-36.6*** (1.5)
Girl × moderator	-22.8*** (1.7)	-0.8** (0.4)	8.0*** (2.8)	9.4*** (2.6)
Child under six in household (N = 34,133)				
Girl	42.5*** (1.1)	0.1 (0.2)	4.1** (1.7)	-35.2*** (1.4)
Girl × moderator	-6.8*** (1.9)	6.4*** (0.7)	-6.3** (3.2)	5.8** (2.7)
Diary day is a Sunday (N = 34,133)				
Girl	38.7*** (1.0)	1.5*** (0.2)	1.6 (1.6)	-31.1*** (1.3)
Girl × moderator	22.7*** (3.3)	1.0 (1.1)	11.0*** (3.6)	-28.0*** (4.5)

Notes: Each block and column is one regression on mixed-sex sets with sibling-set fixed effects, single-year age effects, girl, the moderator and girl x moderator. Girl is the gap where the moderator is zero. The moderators other than the diary day are constant within the set. Mother's education is available where a unique mother is linked. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 reports how the gap varies with household characteristics. The domestic-work gap in urban households differs from that in rural households by -16.1 minutes (SE 2.2), and in the top two consumption quintiles from that in the bottom three by -11.1 minutes (SE 2.0). The gap is present in urban and in richer households, so it is not only a response to poverty or to the demands of rural household production. In this respect it differs from the excess mortality of girls documented by Rose (1999), which responds to income shocks.

The gap is 48.8 minutes (SE 1.2) in sets whose mother has less than secondary education, and the difference for mothers with secondary education or more is -32.2 minutes (SE 1.6). For the household head's education the corresponding difference is -22.8 minutes (SE 1.7). Mothers' education is associated with a smaller division of domestic work between their daughters and sons, consistent with the transmission of attitudes from mothers to children documented by Dhar et al. (2019).

Where a child under six lives in the household, the caregiving gap changes by 6.4 minutes (SE 0.7). Older girls are drawn into the care of young children when there are young children to care for,

which is the sibling-care channel through which a mother's fertility can affect her daughters' time.

5.7 School days and Sundays

On Sunday diaries, the domestic-work gap differs from that on other days by 22.7 minutes (SE 3.3) and the learning gap by 11.0 minutes (SE 3.6). Estimated separately (Table 6), the domestic-work gap is 38.9 minutes (SE 1.0) on Monday to Saturday and 59.9 minutes (SE 3.2) on Sunday. The gap is larger on Sundays, when school does not limit the time available, which suggests that school attendance protects girls' time on school days and that girls' domestic load expands to fill the time that school would otherwise occupy. The survey also asks whether the diary day was a normal routine day. Restricting the sample to normal days gives the estimates in Table 6.

6 Fertility-stopping selection and robustness

Table 6: Robustness and fertility-stopping selection: girl minus boy, minutes per day

Specification	Domestic	Care	Learning	Free time	N
Baseline	40.9*** (1.0)	1.6*** (0.2)	2.6* (1.5)	-33.8*** (1.2)	34,133
Adding age-rank indicators	40.7*** (1.0)	1.6*** (0.2)	2.9** (1.5)	-33.7*** (1.2)	34,133
Adding day-of-week indicators	40.9*** (1.0)	1.6*** (0.2)	2.7* (1.5)	-33.8*** (1.2)	34,133
Siblings report the same diary day	40.9*** (1.0)	1.6*** (0.2)	2.7* (1.5)	-33.8*** (1.2)	34,128
Children of the head only	44.3*** (1.1)	1.7*** (0.2)	2.0 (1.6)	-35.3*** (1.3)	28,944
Two-child sets	36.2*** (1.1)	1.6*** (0.3)	3.5* (1.8)	-34.0*** (1.5)	18,376
No co-resident sibling aged 18+	36.4*** (0.9)	1.4*** (0.2)	3.3** (1.5)	-33.0*** (1.3)	27,897
Eldest child is a boy	38.3*** (1.7)	1.4*** (0.4)	-9.2*** (2.7)	-33.4*** (2.3)	14,923
Eldest child is a girl	20.9*** (1.4)	0.8* (0.4)	8.0*** (2.5)	-23.0*** (2.0)	18,794
Self-reported diaries only	61.3*** (2.3)	1.6*** (0.4)	3.2 (3.3)	-48.9*** (2.8)	6,860
Monday–Saturday diaries	38.9*** (1.0)	1.5*** (0.2)	1.9 (1.6)	-31.5*** (1.3)	30,871
Sunday diaries	59.9*** (3.2)	2.7** (1.1)	8.5*** (3.2)	-54.5*** (4.2)	3,260
Unweighted	39.0*** (0.8)	1.5*** (0.2)	3.1** (1.3)	-32.8*** (1.1)	34,133
Normal routine days only	32.9*** (0.9)	1.0*** (0.2)	0.2 (1.7)	-23.5*** (1.2)	24,957

Notes: Sibling-set fixed effects and single-year age effects. Subsample rows re-define mixed-sex sets within the subsample. Eldest child: the unique oldest child aged 6–17 in the set. Self-reported: both the girl and the boy reported their own diary. N: children in the domestic-work regression. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The sibling sets in the data bear the marks of son-biased fertility. Among all children aged 6 to 17, there are 111 boys for every 100 girls (weighted). Girls live in sibling sets that are 0.17 children larger than boys' (SE 0.01). This is the pattern a son-preferring stopping rule predicts (Yamaguchi 1989; Basu and de Jong 2010).

Table 6 reports the robustness checks. All 13 alternative specifications give a girl-minus-boy domestic-work gap of the same sign as the baseline and significant at the 5 per cent level, ranging from 20.9 to 61.3 minutes; 4 lie within the 95 per cent confidence interval of the baseline. In two-child sets, the domestic-work gap is 36.2 minutes (SE 1.1). In sets whose eldest child is a boy it is 38.3 minutes (SE 1.7), and in sets whose eldest child is a girl 20.9 minutes (SE 1.4). The gap remains large and

significant where a girl has an older brother, the configuration that a stopping rule cannot produce by continuing childbearing in pursuit of a first son, although it is smaller than the baseline. Part of the baseline gap may reflect the birth-order structure created by fertility stopping, but most of it does not. For the head's own children only, which excludes sets that may contain cousins, the gap is 44.3 minutes (SE 1.1). In sets with no co-resident sibling aged 18 or older, it is 36.4 minutes (SE 0.9).

Restricting the sample to sets in which both a girl and a boy reported their own diary gives 61.3 minutes (SE 2.3). The estimate on self-reported diaries departs from the baseline, so proxy reporting may contribute to the measured gap, or self-reporting children may differ from others. Adding child-level day-of-week indicators gives 40.9 minutes (SE 1.0), and restricting the sample to sets in which all siblings report the same diary day gives 40.9 minutes (SE 1.0). The unweighted estimate is 39.0 minutes (SE 0.8).

These checks address the selection that operates within the observed sibling set. They cannot address the selection created by which girls were born and survived, or by girls who married or moved before the survey. If the girls who are missing from the sample are those whose families most disfavoured daughters, the remaining girls come from relatively less discriminating families, and the estimated gaps would understate the gap that would be observed if every girl had been born and stayed at home. Sex selection, as Hu and Schlosser (2015) argue for nutrition, can improve the average circumstances of the girls who are born.

7 Mothers' time and the sex composition of children

Table 7: Mothers' time and the sex composition of their co-resident children

Mother's minutes per day	All daughters (6–17)	Daughters 10–17 (vs sons)	Mean
Active childcare	2.7*** (0.8)	1.3** (0.5)	54.4
All childcare	3.2*** (0.8)	1.7*** (0.5)	56.9
Unpaid domestic work	-7.7*** (1.4)	-9.8*** (1.2)	365.6
Employment	5.3** (2.1)	8.8*** (1.8)	102.6

Notes: Linked mothers aged 20–49 with at least one co-resident child aged 6–17 (N = 45,732). Column 1: indicator that all such children are daughters. Column 2: number of daughters aged 10–17, controlling for the number of children aged 10–17 (so the coefficient compares a daughter with a son). Controls: mother's age band and education, urban, consumption quintile, number of children aged 6–17, members under six, employment (except for the employment outcome), proxy report, and state and day-of-week fixed effects. FSU-clustered standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

The sibling comparison holds the parents fixed, so it cannot show whether parents' own time depends on the sex of their children. Table 7 compares mothers whose co-resident children aged 6 to 17 are all daughters with mothers who have at least one such son, conditional on the number of children and household characteristics. Mothers of only daughters do 2.7 minutes (SE 0.8) more active childcare and 3.2 minutes (SE 0.8) more childcare in total. Their domestic work differs by -7.7

minutes (SE 1.4) and their employment time by 5.3 minutes (SE 2.1).

The second column asks whether daughters substitute for their mothers' domestic work. Conditional on the number of children aged 10 to 17, each adolescent daughter in place of an adolescent son is associated with 9.8 minutes (SE 1.2) less domestic work by the mother. Mothers with adolescent daughters do less domestic work, which is consistent with daughters taking over part of their mothers' load.

These estimates lack a household fixed effect. Under son-biased stopping, mothers with only daughters differ from others in ways that may be related to their time use: they may have more children, may be more likely to be still having children, and may be in families with weaker or stronger son preference. I control for the number of children, the mother's age and household characteristics, but treat these results as secondary evidence.

8 Discussion

8.1 Interpretation

Daughters in India do more of the household's domestic work than their brothers of the same age in the same household, and the sibling design rules out explanations based on differences between families. What the design does not settle is why. Three explanations fit a within-family gap.

The first is son preference in the narrow sense: parents value sons' well-being or future more and allocate scarce study and leisure time accordingly. Patrilocal marriage, under which daughters leave the household and sons stay, gives parents a reason to invest more in sons' schooling (Das Gupta et al. 2003; Kingdon 2005). The second is a norm about tasks that is independent of how much parents value each child: girls cook and clean because that is what girls do, and identity costs attach to departures from it (Akerlof and Kranton 2000; Jayachandran 2015). The third is specialisation on expected comparative advantage: if parents expect daughters to run a household as adults and sons to work for pay, training daughters in domestic tasks may look like an investment in the daughter rather than a cost to her (Becker 1991; Pitt et al. 1990).

The evidence in this paper speaks to these explanations only indirectly. That girls' learning time is not significantly lower than their brothers' means that the domestic burden competes mainly with free time and rest rather than with education. The presence of the gap in urban and richer households, where the household's demand for children's work is plausibly lower, points to norms or preferences rather than to scarcity alone. The additional load on eldest daughters fits a model in which daughters substitute for mothers when there is work to be done. A formal test between the explanations would require variation in the returns to girls' education or in norms, such as the variation used by Jensen (2012) and Beaman et al. (2012).

8.2 Policy

Programmes that aim to keep girls in school, such as conditional transfers or the bicycle programme studied by Muralidharan and Prakash (2017), measure success by enrolment and attendance. The results here suggest also measuring what happens to girls' time outside school. If a girl's domestic load is fixed by the household's needs and by norms, attending school may reduce her free time and rest rather than her domestic work. Interventions that work on norms in adolescence, such as the school curriculum studied by Dhar et al. (2022), can target the division of household tasks between sisters and brothers directly, and girls' domestic workload is a natural outcome measure for them. The official SDG indicator on unpaid domestic and care work is reported by sex and age, and the national time-use reports (National Statistics Office 2025) can report children's unpaid work separately so that the age at which the adult gap opens is visible in official statistics.

8.3 Limitations

The survey records one diary day per person. A single day measures the average day in a population well but measures each child's usual day with error. Because sisters and brothers report the same day in almost all sets, the error is common to them and is absorbed by the fixed effect when it reflects the day itself, but not when it reflects child-specific variation. Children under six keep no diary, so the analysis starts at age six, after some of the processes that shape girls' health and nutrition have already run their course. Children who live away from their parents, including girls who married before 18 and children sent to relatives or to boarding schools, are not observed. The survey does not record which child a parent's care is directed to, so the parental analysis relies on sex composition and cannot be estimated within the family. Finally, I do not examine regional differences. The north–south contrast in son preference (Dyson and Moore 1983; Miller 1981) is a natural dimension of heterogeneity, but the data dictionary available for this study does not document the state codes, and I do not assign states to regions without it. State fixed effects are absorbed by the sibling-set fixed effect.

9 Conclusion

Most evidence on son preference in India measures what parents give to sons and to daughters. This paper measures what households ask of them. Comparing sisters and brothers aged 6 to 17 in the same household with full-day diaries from the Time Use Survey 2024, I find that girls do 40.9 minutes (SE 1.0) more domestic work and 1.6 minutes (SE 0.2) more caregiving than their brothers, with 2.6 minutes (SE 1.5) more time in learning and 33.8 minutes (SE 1.2) less free time. The domestic-work gap widens through adolescence. It is smaller but still large and significant in two-child sets and in sets with an elder brother, so it is not mainly an artefact of fertility stopping. The time a girl spends on household work is a measure of gender discrimination that can be collected in any time-use survey, compared across countries and survey rounds, and used to evaluate programmes aimed at adolescent girls.

Data and code availability

The Time Use Survey 2024 unit-level data are distributed by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India (microdata.gov.in, catalogue 236), to registered users. Code available from the author on request; will be made public on acceptance.

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Appendix

Table 8: Construction of the analysis samples

Step	N
Diaries in the file	450,408
Children aged 6–17, never married, children or grandchildren of the head	82,223
In a set with at least two children aged 6–17	54,484
In a mixed-sex set (fixed-effects sample)	34,133
Mixed-sex sets	13,984
Mixed-sex sets with a unique eldest child	13,825
Children with a linked mother	31,832
Linked mothers aged 20–49 (mothers' analysis)	45,732

Notes: A sibling set is the children with the same relationship to the head (child or grandchild) in the same household.

Table 9: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.